

CAMPRISA Reviewer

October 4, 2025

This document holds most of the content that we have learnt over the while of training. I regretfully admit my lacking of understanding to many topics because the mathematics that I have learnt from these training sessions are of unknown territory, but I have some time trying to comprehend. Considering how I have I have tried my best to understand what I have learnt from these topics, so if any errata persist you can always contact me. SOME TOPICS ARE *OMITTED* FOR SIMPLICITY AND TIME SAKES.

These notes are listed on my website linked here. and the direct link to the collection of notes I have made is through this hyperlink. I will also provide the source code of the $\text{\LaTeX}$ file after I upload it to my GitHub repository.

Made with love and $\text{\LaTeX}$
Made in $\text{\TeX}$ Maker and Neovim
 $\text{\LaTeX}$ distribution used: Mik $\text{\TeX}$

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1 Arranging Objects

1.1 The Fundamental Principle of Counting

Example.

Charles has three teddy bears; colored brown, white, and grey. She bought five small hats (blue, black, brown, grey, and red colored) that go alongside the teddy bears. How many possible pairs of hats and teddy bears can there be?

We note that we consider it a *pair*, so a hat cannot go atop one already worn. So three times five yields *fifteen possible pairs*.

Each bear can only wear one hat, and hence each bear is only restricted to five options and not less (a bear wearing two hats leaving three other options left.)

This thorough explanation is an example of the *fundamental principle of counting*. We can represent this as a "tree" or "network" of items, like in this following example.

Example.

Liam is in a restaurant. They serve three main course dishes, three side dishes, and two drinks. What are their possible orders? Consider that:

1. They serve: Chicken, Beef, and Pork.
2. Their side dishes can be: Soup, Fries, and Garlic Bread.
3. Their drinks are: Water and Iced Tea.

There are multiple options since they can branch out. We can create a network (a tree) that represents the possible choices; starting with the main course as the start.

Checking below,

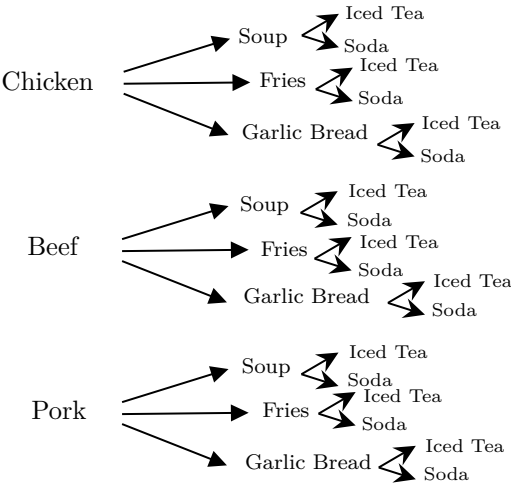


Figure 1: From The Previous Example, Illustrated Awfully

Counting all possible orders, we get *eighteen possible orders*. We can also check by multiplying the number of possible options together $3 \cdot 3 \cdot 2 = 18$.

Example.

A survey has ten questions with two answers per question. How many possible results can there be?

We know that when we answer a question, it does not affect any other question. It can be all Option A or half Option A and Option B. Meaning, we just need to multiply all of the potential results together.

We can do the tree approach and list every possible answer for each question, but that would take forever. We can just multiply all possible options (two answers *per* question) together. We get 2^{10} , therefore *there are 1024 possible results*.

Definition 1.1 (Fundamental Principle of Counting). If an event occurs in a_1 ways, then the following events occur in $a_2, a_3, \dots$ ways respectively, then the amount of potential events is in $a_1 \cdot a_2 \cdot a_3 \cdot \dots$ ways.

Remark.

If an event A does not directly change the outcome of the succeeding event B , then the events are considered *independent*. If an event A does directly change the outcome of the succeeding event B , then the events are considered *dependent*.

Example 1 is considered dependent since if an option has been chosen the other options can not be chosen anymore.

Examples 2 and 3 are considered independent because: each order from the restaurant is different and it does not imply that when I pick soup over fries I can't order fries anymore, and that the survey's questions do not influence the next one.

1.2 Factorial Notation

Definition 1.2. A factorial of a number n represented as $n!$ is expressed as:

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 3 \cdot 2 \cdot 1$$

Remark.

To simplify factorials as a rational expression: Using the definition of a factorial,

$$\frac{n!}{m!(n-1)!} = \frac{n(n-1)(n-2)\dots 3 \cdot 2 \cdot 1}{m! \cdot (n-1)(n-1-1)(n-2-1)\dots 3 \cdot 2 \cdot 1}$$
$$\frac{\cancel{n(n-1)(n-2)\dots 3 \cdot 2 \cdot 1}}{m! \cdot \cancel{(n-1)(n-2)(n-3)\dots 3 \cdot 2 \cdot 1}} = \frac{n}{m!}$$

If there are any terms in the numerator and denominator when expanding the factorial term as a series of products, you can cancel identical terms seen on both factorials.

Example.

$$\begin{aligned} \frac{3!}{5!} &= \frac{3 \cdot 2 \cdot 1}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} & (1) \\ \frac{\cancel{3 \cdot 2 \cdot 1}}{5 \cdot 4 \cdot \cancel{3 \cdot 2 \cdot 1}} &= \frac{1}{5 \cdot 4} & (2) \\ &= \frac{1}{20} & (3) \end{aligned}$$

1.3 Permutation

A *permutation* is the set of all possible re-arrangements of a certain group of elements wherein order matters (meaning $1, 2, 3 \neq 2, 1, 3$).

Definition 1.3. A set of n objects taken n at a time:

$$P(n, n) = n!$$

$P(n, n)$ can be written as ${}_nP_n$ or P_n^n .

Example.

How many ways can you arrange letters q,r,t,u,a,g?

Count there are six letters total. We want to arrange it to form a group of six. We input that to our permutation,

$$P(6, 6) = 6! = \boxed{720}$$

Definition 1.4. A set of m objects taken n at a time:

$$P(m,n) = \frac{m!}{(m-n)!} \text{ for } m > n$$

Example.

Lorine’s laptop has a five-character PIN using digits 0 to 9. In how many ways could we arrange numbers for Lorine’s PIN?

Infer that there are ten single-digits from 0 to 9.

$$\begin{aligned} P(10,5) &= \frac{10!}{(10-5)!} \\ \frac{10!}{5!} &= \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot \cancel{5!}}{5!} \\ \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot \cancel{5!}}{\cancel{5!}} &= 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \\ &= \boxed{30240 \text{ ways.}} \end{aligned}$$

Definition 1.5. A set of objects with identical elements has a permutation enumerated as:

$$P = \frac{\delta!}{a!b!c! \dots}$$

Wherein δ is the amount of objects, and the denominator is the amount of each identical object.

Example.

In the word SKIBIDI, how many ways can you arrange its letters?

There is only one set of identical letters, three letter I’s.

$$\begin{aligned} P &= \frac{7!}{3!} \\ \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3!}{3!} &= \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot \cancel{3!}}{\cancel{3!}} \\ 7 \cdot 6 \cdot 5 \cdot 4 &= \boxed{840} \end{aligned}$$

Definition 1.6 (Circular Permutation). A set of n distinct objects put on the side of a circle has a permutation of:

$$P = (n-1)!$$

I personally would not recommend looking up the circular permutation for identical objects. Can’t even consider it High School maths.

Example.

On the side of a table are five chairs. If me, Liam, Charles, Lorine, and Sir Rodel were to sit down on those chairs, how many ways can we sit down on each chair?

$$(5-1)! = 4! = \boxed{24 \text{ ways.}}$$

Remark.

- Possible key-terms when scanning the question if it asks for permutation:
1. "rearrange"
 2. "arrangement"
 3. "ordered ways"
 4. "unique ways"
 5. "distinct ways"
 6. Questions involving selecting from a group of people as President, Vice Pres., Secretary, etc.
 7. "as a code"
 8. "permutation"
 9. "... order matters"
 10. "line up"
 11. "positioned"
 12. "ranked"
 13. Any question that involves a password unless specified otherwise.

1.4 Combination

A *combination* is the set of all possible rearrangements of a certain group of elements wherein order *does not matter*.

Example.

For there are three items, find all possible groups of two.

Say there are items A, B, C . We can find AB, AC, BC are groups. However, we cannot include the group BA because it is the same group as AB .

Hence there are $\boxed{3}$ possible groups.

A set of m objects chosen n at a time:

$$C(m, n) = \frac{m!}{n!(m - n)!} \text{ form } m > n$$

Example.

I have a list of six items our group needs to buy for Teenpreneur. I want to sort them by groups of two so I can give it to my other two teammates. How many ways can I group these items?

$$\begin{aligned} C(6, 2) &= \frac{6!}{2!(6 - 2)!} \\ \frac{6!}{2!4!} &= \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2!}{2!4!} \\ \frac{6 \cdot 5 \cdot \cancel{4} \cdot \cancel{3}}{\cancel{4} \cdot \cancel{3} \cdot 2 \cdot 1} &= \frac{6 \cdot 5}{2} \\ &= \boxed{15 \text{ ways.}} \end{aligned}$$

SIDE NOTE: Teenpreneur is an annual (I think?) event in our school wherein students are grouped into teams and they sell food at a stall.

Remark.

- Possible key-terms when scanning the question if it asks for combination:
1. "select"
 2. "choose"
 3. "pick"
 4. "teams"
 5. "handshakes"
 6. "combination"
 7. "partition"
 8. "divide"
 9. "... order doesn't matter"
 10. "group"
 11. "subset"

BE WEARY! Read the question *carefully* because some terms might overlap. For example, a *combination lock* is a permutation.

2 Probability

Probability is the study on the chance of an event occurring given a set of possible outcomes. This topic is *quite* diverse, you can tell because a graduate-level course titled "Probability Theory" exists.

I will assume that the reader knows about *set theory notation*. I have my reasons I won't append it in the appendix but you can research or use the book from Grade 7.

2.1 Introduction

The *odds* of an event are the potential outcomes that are meant favorable. It is the ratio between all favorable outcomes and all possible outcomes.

$$\text{Odds} = \frac{\text{favorable outcomes}}{\text{possible outcomes}}$$

The *sample space* (formally S) is the set of all possible outcomes. We will only deal with finite sample spaces.

Example.

A coin is tossed twice. Find the probability that the landing will be both tails.

- There are two favorable outcomes: (T, T) .
- There are four potential outcomes: $(H, T), (H, H), (T, H), (T, T)$.

Call the set of favorable outcomes P and the set of possible outcomes Q or the sample space. Find their cardinality.

$$\text{Card}(P) = 1$$
$$\text{Card}(Q) = 4$$

As a ratio, there is a $\frac{1}{4}$ chance that tails is landed twice.

Let us try another example.

Example.

Two die are tossed. What is the probability that the potential die's landing is a sum that equals five?

There are six faces each die. A quick way to find all possible outcomes is multiplying all faces: 36.

There are four possible sums of five:

- $(1, 4)$
- $(2, 3)$
- $(3, 2)$
- $(4, 1)$

$$\frac{\text{Card}(P)}{\text{Card}(Q)} = \frac{4}{36}$$
$$= \frac{1}{9}$$

Therefore, there is a $\frac{1}{9}$ chance it is a sum of five. We shall redefine the Card A as $P(A)$ or "the results favorable/possible of A ."

2.2 Compound Events

What we talked about are simple events. They only have one event going on. A *compound event* has two or more events occurring. They can be *exclusive* or *joint* events.

- An event is *exclusive* (mutually exclusive in our book) if events happening can not occur at the same time; drawing a number card and a king.
- An event is *joint* (non-mutually exclusive) if events happening do occur at the same time; rolling one dice that makes one a two and another an even number.

Definition 2.1. The union of all favorable outcomes in exclusive events A and B as $P(A \cup B)$ is represented as:

$$P(A \cup B) = P(A) + P(B)$$

This is read as " A or B ."

Example.

Find the probability of *either* drawing a number two card and a queen.

There are four number two cards in a standard deck of 52 (set A). You can get a head from two faces of a coin (set B).

$$P(A \cup B) = P(A) + P(B)$$
$$\frac{4}{52} + \frac{1}{2} = \frac{4}{52} + \frac{26}{26} \cdot \frac{1}{2}$$
$$\frac{4 + 26}{52} = \boxed{\frac{15}{26} \text{ chance.}}$$

Definition 2.2. The union of all favorable outcomes in joint events A and B as $P(A \cup B)$ is represented as:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

This is still read as " A or B ." The intersection is meant for both events occurring.

Example.

Find the probability of tossing a dice and getting a factor of six or a multiple of two.

There are four faces that give a factor of six (set A). There are three faces that give a multiple of two (set B). There are exactly two faces that both give a multiple of two and is a factor of six (set $A \cap B$).

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ \frac{4}{6} + \frac{3}{6} - \frac{2}{6} &= \frac{1}{3} + \frac{1}{2} - \frac{2}{6} \\ \frac{4 + 3 - 2}{6} &= \boxed{\frac{5}{6} \text{ chance.}} \end{aligned}$$

2.3 Independent and Dependent Events

An event is *dependent* if the first event changes the outcome of the second event. An event is *independent* if no event changes any outcome of the succeeding.

Definition 2.3 (Independent Events). The intersection of all favorable outcomes in events A and B as $P(A \cap B)$ is represented as:

$$P(A \cap B) = P(A) \cdot P(B)$$

Read as " A and B "

Example.

Find the probability of landing heads and getting an even number on one dice.

$$\begin{aligned} P(A \cap B) &= P(A) \cdot P(B) \\ \frac{1}{2} \cdot \frac{1}{2} &= \boxed{\frac{1}{4} \text{ chance.}} \end{aligned}$$

Definition 2.4 (Dependent Events). The intersection of all favorable outcomes that change in correlation to event A and to B as $P(A \cap B)$ is represented as:

$$P(A \cap B) = P(A) \cdot P(B|A)$$

$P(B|A)$ is the second event that is a *reduced* amount of potential outcomes of B because of event A .

Illustrating as an example,

Example.

Liam draws an Ace of Club, and Charles draws a Queen of Hearts. What is the probability of this occurring?

Once Liam draws an Ace of Club, he *does not return it back*. Therefore, the sample space is one less card. Therefore, Charles just needs to draw 51 cards instead of 52 to have a chance of a Queen of Hearts.

Drawing an Ace of Club is a $\frac{1}{52}$ chance. Now that Liam got the card, Charles drawing a Queen of Hearts is $\frac{1}{51}$ chance.

$$\begin{aligned} P(A \cap B) &= P(A) \cdot P(B|A) \\ \frac{1}{52} \cdot \frac{1}{51} &= \boxed{\frac{1}{2652} \text{ chance.}} \end{aligned}$$

Incredibly slim chances, huh?

Remark.

Whenever the question has a phrase that defines the same as *not returning an item back*, it is dependent. If the question does mention *returning it back*, it is independent.

3 Basic Set Theory

3.1 Element of, Union, Intersection

A *set* as described earlier is a collection of objects called *elements*. Elements can be repeated.

$$A = \{a, b, c, d, \dots\}$$

Sets can be infinite or finite. If an element is a part of a set, we use the symbol $\in$.

$$a \in B$$

given $B = \{1, a, 2, b, c\}$.

Definition 3.1 (Union of Two Sets). Given two sets A and B , the *union* of both sets $A \cup B$ is the set of all possible elements in each sets, with elements that exist in both sets only appearing once.

Example.

Given a set of $A = \{2, 4, 6, 9\}$ and $K = \{1, 4, 5, 9, 10\}$, $A \cup B$ is:

$$A \cup K = \{1, 2, 4, 5, 6, 9, 10\}$$

Union lists all elements that exist in either a single set or in both. This means no element can be repeated under a union set.

Definition 3.2 (Intersection of Two Sets). Given two sets A and B , the *intersection* of both sets $A \cap B$ is the set of all common elements only from both sets.

The intersection of the sets A and K from earlier:

$$A \cap B = \{4, 9\}$$

3.2 Relative Complement, Universal Set, and Universal Complement

Definition 3.3 (The Relative Complement). The *relative complement* of two sets A and B is the list of elements that are in A but not in B , $A \setminus B$.

For sets A and K from earlier,

$$\begin{aligned} A \setminus K &= \{2, 4\} \\ K \setminus A &= \{1, 5, 10\} \end{aligned}$$

If both sets have no elements that are common of each other, the sets are *disjoint*. Say $P = \{a, b, c\}$ and $Q = \{1, 2, 3\}$.

$$\begin{aligned} P \setminus Q &= \{a, b, c\} \\ Q \setminus P &= \{1, 2, 3\} \end{aligned}$$

Definition 3.4 (The Universal Set). Given an n amount of sets, there exists a set U that houses all elements of each set, including duplicates. This set is called the *universal set* U .

Example.

Given ALL sets listed earlier; $A = \{2, 4, 6, 9\}$, $K = \{1, 4, 5, 9, 10\}$, $P = \{a, b, c\}$, and $Q = \{1, 2, 3\}$, the universal set U is:

$$U = \{1, 1, 2, 2, 3, 4, 4, 5, 6, 9, 9, 10, a, b, c\}$$

Definition 3.5. Given there exists a universal set U and a set A , the *universal complement* A' is the list of all elements in A that are not in the universal set.

The set $R = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ has the universal complement:

$$R' = \{7, 8\}$$

3.3 Subsets

Simply put, a set is considered a *subset* if all elements are in another set. If only a certain amount of elements in a set are also elements of the other set, it is considered a *proper subset*. Otherwise, it is an *improper subset*.

Definition 3.6. If two sets A and B are proper subsets, we use $A \subset B$. If they are improper subsets, then $A \subseteq B$.

Example.

Every set is a subset of the universal set, $K \subset U$. Every set is an improper subset of itself, $A \subseteq A$. The union of two sets has always the separate sets as a subset; given A and K from earlier,

$$A \subset A \cup K \text{ and } K \subset A \cup K$$

3.4 The Null Set

Definition 3.7. The *empty set* or the *null set* $\emptyset$ is a set with no elements.

If two sets form an intersection despite having no common elements, it results to an empty set. The empty set is a subset of all sets, and a subset of itself. But no set is a subset of the null set.

4 Coordinate Geometry Formulas

Definition 4.1 (Distance Formula). Given a line $\overline{AB}$ with points A and B on the cartesian plane with coordinates $A = (x_1, y_1)$ and $B = (x_2, y_2)$, the *distance* from each point on a line is determined by:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Definition 4.2 (Midpoint Formula). given a line $\overline{AB}$ with the same coordinates as prior, the midpoint coordinate of the line can be determined by:

$$m = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

5 Basic Trigonometry

5.1 On Right Triangles

You can read more [here](#), Just ignore the final section.

5.2 Trigonometry on the Unit Circle

See the following diagram of a circle.

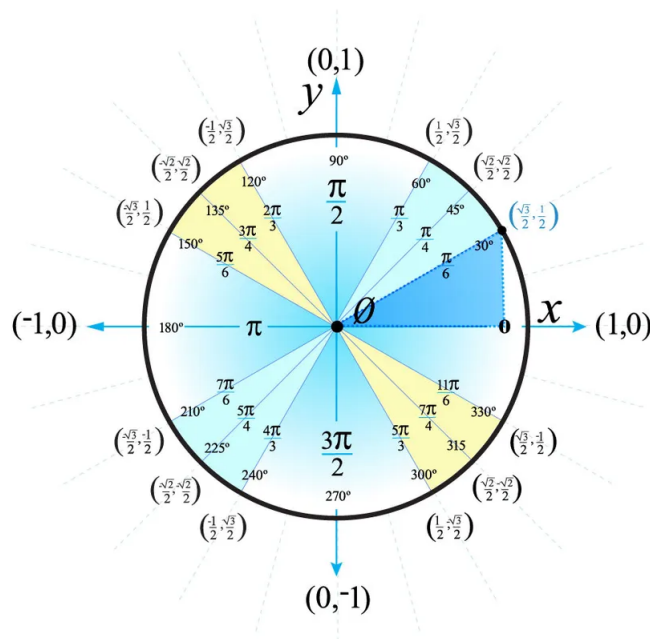


Figure 2: Unit Circle on the Cartesian Plane

6 Mean, Median, and Mode

Definition 6.1 (Mean). The *mean*, or *average*, is the sum of all a list of numbers divided by how many there are.

$$\frac{\text{sum of all values}}{\text{amount of values}}$$

or using mathematics: for a group of numbers $a_i = a_1, a_2, a_3, \dots$ for $i \in \mathbb{Z}$,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n a_i$$

Wherein i is the index for the list of numbers (not really important), n is the amount of values there are, and a_i are the numbers themselves. $\bar{x}$ is typically the symbol for a mean.

Definition 6.2 (Median). The *median* is the middle number of a list of numbers. If there are an even numbered amount of numbers, we divide the two middle numbers by two to get the median of that list.

Definition 6.3 (Mode). The *mode* is the most frequent number. If there are n numbers that appear the same amount of frequency, there are n modes. If numbers are all distinct, then there exists no mode.

7 Rate and Proportion

A *rate* is the comparison of two quantities with different units.

$$a : b = c : d$$

b and c are the *means* whilst a and d are the *extremes*.

Remark.

When finding an inverse proportion related to distance, we note the numbers. r_0 and r_1 are the smaller and bigger rates respectively, and t_0 and t_1 are the smaller and bigger time allocated respectively.

$$\frac{r_0}{r_1} = \frac{t_0}{t_1}$$

Definition 7.1 (Formal Definition of Partitive Proportion). Given a set of parts $p_i = p_1, p_2, p_3, \dots, p_n$, we define the partitive proportion as $p_1 : p_2 : p_3 : \dots : p_n$, The total amount

divided per group is as:

$$q = n \sum_{i=1}^n p_i$$

or the product of the sum of all parts and the amount of parts that will be distributed to a specified amount per group.

Hard to explain.

Example.

Divide 48 pencils to children in the matter where 1 : 1 : 2.

$$\begin{aligned} 48 &= (1 + 1 + 2)n \\ 48 &= 4n \\ n &= 12 \end{aligned}$$

Divide 12 pencils by how many is specified per group. Two children just want one set of the partitioned pencils, whilst the selfish one wants double; $1 \times 12 : 1 \times 12 : 2 \times 12 = 12 : 12 : 24$.

8 Percent

8.1 Percentage, Base, Rate

Use the PBR triangle:

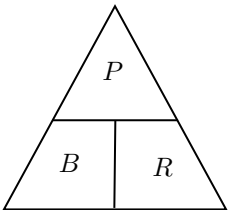


Figure 3: The PBR Triangle

$$\begin{aligned} P &= BR \\ B &= \frac{P}{R} \\ R &= \frac{P}{B} \end{aligned}$$

Remark.

In the statement "n% of m is ...", n% is the percent per 100, "of" is multiplying to the next number which is replaced with a set of ellipses.
P is the percentage acquired (after "is"), B is the literal percentage, and R is the base value.

8.2 The Original Price of a Discounted Product

Recall that to find the discounted price of a product p by n%, it is simply $P_{discount} = p - pn\%$, wherein n% is expressed as a decimal. To find the original price. To find the original price,

$$\begin{aligned} P_{discount} &= p - pn\% \\ P_{discount} &= p(1 - n\%) \\ p &= \frac{P_{discount}}{1 - n\%} \end{aligned}$$

This is the formula for obtaining the original price.

$$p = \frac{P_{discount}}{1 - n\%}$$

9 The Distance, Rate, Time Triangle

Note in a similar triangle to the PBR Triangle, the DRT Triangle:

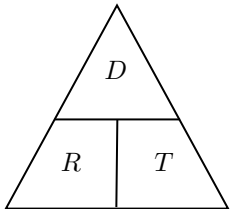


Figure 4: The DRT Triangle

$$\begin{aligned} D &= RT \\ R &= \frac{D}{T} \\ T &= \frac{D}{R} \end{aligned}$$

Example.

An object runs at 4 meters a minute. The object ran a distance of thirty-six meters. How long (in minutes) did it take the object to fully traverse the distance?

$$T = \frac{D}{R} = \frac{36}{4} = 9 \text{ minutes}$$

10 Geometry

10.1 Area and Perimeter of Geometric Shapes

Illustration is not given. NOTE THAT: if the question does not specify an approximation for π , immediately assume to treat π as a constant. Typically, $\pi = 3.14159265389\dots$ or $\pi = 3.14$.

PERIMETER

- | | |
|--|--|
| 1. Square: $4a$ for all equal sides a | 6. Parallelogram: $2(b + h)$ for the base b and height h |
| 2. Triangle: $a + b + c$ for all sides a, b, c | 7. Circle: $2\pi r$ for a radius r |
| 3. Rectangle: $2(l + w)$ for shorter side w and longer side l | 8. Isosceles Triangle: $2a + b$ since two sides are equal $a = b$ and only two |
| 4. Equilateral Triangle: $3a$ since all sides are equal, $a = b = c$ | 9. Ellipse: don't even think about it. |
| 5. Trapezoid: $b_1 + b_2 + l_1 + l_2$ for both bases b_1, b_2 and remaining sides l_1, l_2 | 10. Rhombus: same as square since equal sides. |

AREA

- | | |
|--|---|
| 1. Square: a^2 | a right triangle on one of the sides. |
| 2. Triangle: $\frac{1}{2}bh$ for base b and height from the middle h | 6. Parallelogram: bh for the base b and height h . |
| 3. Rectangle: lw | 7. Circle: πr^2 |
| 4. Equilateral Triangle: $\frac{\sqrt{3}}{4}a^2$ (this is not included) | 8. Isosceles Triangle: same as a triangle |
| 5. Trapezoid: $\frac{h(b_1 + b_2)}{2}$ for height h forming | 9. Ellipse: $\pi r_1 r_2$ for any two radii r_1, r_2 |
| | 10. Rhombus: $\frac{pq}{2}$ given a diagonal p and q traced from opposite vertices of the rhombus |

10.2 Surface Area and Volume of Geometric Solids

Geometric Solid	Volume	Surface Area	Information
Cube	s^3	$6e^2$	for any side s , for any edge e
Rectangular Prism	lwh	$2(lw + lh + wh)$	height h , length l , width w
Cylinder	$\pi r^2 h$	$2\pi r(r + h)$	for any radius r and height h
Cone	$\frac{1}{3}\pi r^2 h$	$\pi r(1 + r)$	for any radius r and height h
Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$	for any radius r

10.3 Theorems on Geometry

For Grade 8, click [here](#) (I was twelve when I made this, weary for mistakes/weird grammar.). For Grade 9, click [here](#). Grade 10 Geometry is typically not covered in CAMPRISA. It is mostly within 8-9.

10.4 Formulas for Polygons

Definition 10.1 (Formulas Related to Angles, Sides, and Diagonals). A *diagonal* is a line traced from an opposite point to another.

- Sum of all angles in an n sided polygon: $180(n - 2)$
- For an angle in a regular polygon: $\frac{180(n - 2)}{n}$
- How many diagonals are in an n sided polygon: $\frac{n(n - 3)}{2}$

And if it wasn't clear, n is the amount of sides.

By default, a triangle has a sum of 180° , a quadrilateral has a sum of 360° , and these are values to be weary of intensely.

10.5 Angles

Refer to your Grade 7 book and Grade 8 book for total overview.

- An *adjacent angle* is an angle that share a common vertex.
- A *vertical angle* is non-adjacent and is formed by two intersecting lines. Two pairs of angles have congruence.
- A *linear pair* are adjacent, but a pair of rays that share a common vertex are opposite each other. They are supplementary.

11 Factoring

Definition 11.1 (Forms of Factoring).

- Factoring out a common term: $ax + ay = a(x + y)$
- Difference of two squares: $x^2 - y^2 = (x + y)(x - y)$
- The square of a binomial:
$$(x + y)^2 = x^2 + 2xy + y^2$$
$$(x - y)^2 = x^2 - 2xy + y^2$$
- Cube of a binomial:
$$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$
$$(x - y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$$
- Sum of two cubes: $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$
- Difference of two cubes: $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$
- By grouping terms: $ax + ay - bx - by = a(x + y) - b(x + y) = (a - b)(x + y)$

How to factor a regular trinomial when the lead coefficient is 1

For a general trinomial $x^2 + (a + b)x + ab$, it is always factorable to $(x + a)(x + b)$. Meaning, find two numbers that add to the coefficient of the second term, and multiply to the third term. Note that signs count.

Example.

1. $x^2 + 5x + 6 = (x + 3)(x + 2)$

2. $x^2 - 5x + 6 = (x - 3)(x - 2)$

3. $x^2 + 5x - 6 = (x - 1)(x + 6)$

4. $x^2 - 5x - 6 = (x - 6)(x + 1)$

The numbers mentioned in the factored form, including the negative/positive sign, add to the middle coefficient and multiply to the constant.

If I did not explain well, there are numerous videos relating to it. For when the leading coefficient is greater than 1, watch [this](#).

12 Miscellaneous

KING HENRY DIED UNUSUALLY DRINKING CHOCOLATE MILK

- 1. **K:** Kilo-
- 2. **H:** Hecto-
- 3. **D:** Deca-
- 4. **U:** Unit measure, or without a prefix
- 5. **D:** Deci-
- 6. **C:** Centi-

Each increment is multiplied by 10 going up, and divided by 10 going down.

Converting Farenheit to Celsius and vice versa:

$$C^{\circ} = \frac{5}{9}(F^{\circ} - 32)$$
$$F^{\circ} = \frac{9}{5}(C^{\circ} + 32)$$

To my teammates: *Good luck.* We had fun. Leave the quadratics, sequence and series, and linear systems to me. Arigato.

To Sir Neil: Thank you for educating us in CAMPRISA. You taught it in a unique style and helped perfect some skills lacking. Wishing you much more in this school year. Arigato.